

AWS Managed AI Services

Why AWS AI Managed Services?

- **AWS AI Services are pre-trained ML services for your use case**
- **Responsiveness and Availability**
- **Redundancy and Regional Coverage:** deployed across multiple Availability Zones and AWS regions
- **Performance:** specialized CPU and GPUs for specific use-cases for cost saving
- **Token-based pricing:** pay for what you use
- **Provisioned throughput:** for predictable workloads, cost savings and predictable performance



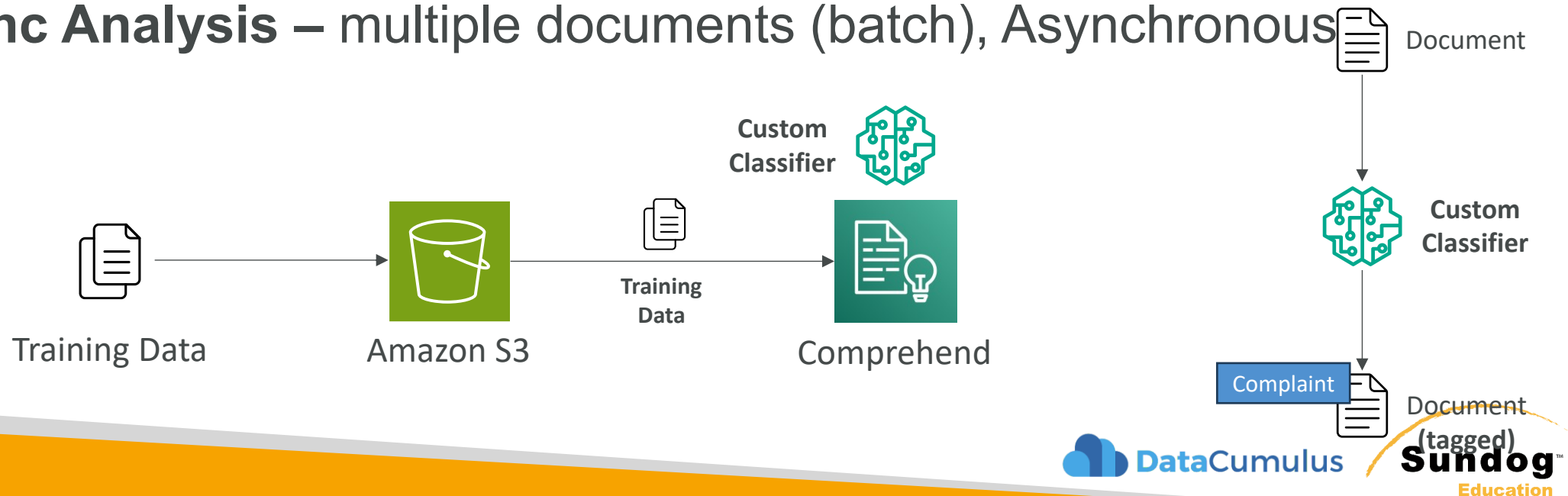
Amazon Comprehend



- For **Natural Language Processing – NLP**
- Fully managed and serverless service
- Uses machine learning to find insights and relationships in text
 - Language of the text
 - Extracts key phrases, places, people, brands, or events
 - Understands how positive or negative the text is
 - Analyzes text using tokenization and parts of speech
 - Automatically organizes a collection of text files by topic
- Sample use cases:
 - analyze customer interactions (emails) to find what leads to a positive or negative experience
 - Create and groups articles by topics that Comprehend will uncover

Comprehend – Custom Classification

- Organize documents into categories (classes) that you define
- Example: categorize customer emails so that you can provide guidance based on the type of the customer request
- Supports different document types (text, PDF, Word, images...)
- **Real-time Analysis** – single document, synchronous
- **Async Analysis** – multiple documents (batch), Asynchronous



Named Entity Recognition (NER)

- **NER** – Extracts predefined, general-purpose entities like people, places, organizations, dates, and other standard categories, **from text**

Hello Zhang Wei, I am John. Your AnyCompany Financial Services, LLC credit card account 1111-0000-1111-0008 has a minimum payment of \$24.53 that is due by July 31st. Based on your autopay settings, we will withdraw your payment on the due date from your bank account number XXXXXX1111 with the routing number XXXXX0000.

Customer feedback for Sunshine Spa, 123 Main St, Anywhere. Send comments to Alice at sunspa@mail.com.

I enjoyed visiting the spa. It was also very expensive. The amenities were ok but the service made the spa a great experience.

Entity 10: 123 Main St ✕

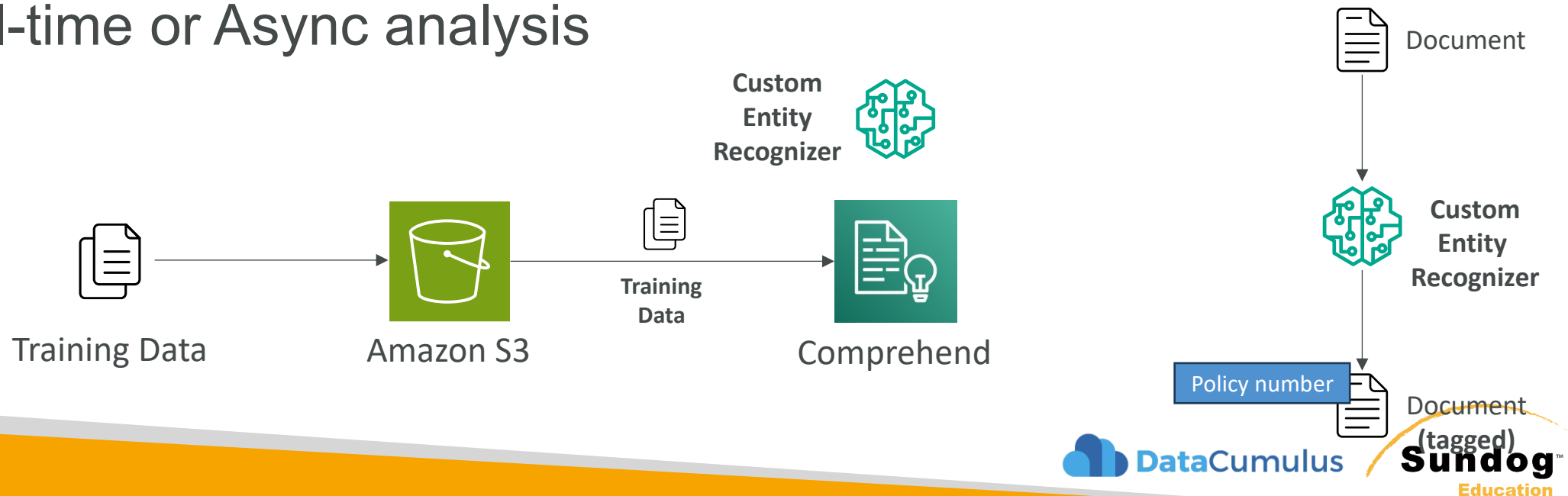
Entity: LOCATION

Confidence: 0.98

Entity ▾	Type ▾
Zhang Wei	Person
John	Person
AnyCompany Financial Services, LLC	Organization
1111-0000-1111-0008	Other
\$24.53	Quantity
July 31st	Date
XXXXXX1111	Other
XXXXX0000	Other
Sunshine Spa	Organization
123 Main St	Location

Comprehend – Custom Entity Recognition

- Analyze text for specific terms and noun-based phrases
- Extract terms like policy numbers, or phrases that imply a customer escalation, anything specific to your business
- Train the model with custom data such as a list of the entities and documents that contain them
- Real-time or Async analysis



Comprehend – Custom Models

- You can create custom models for entity recognition or document classification
 - Trained on your own data
- Comprehend manages your model versioning
- Custom models may be copied between AWS accounts
 - Attach IAM policy to a model version, authorizing the other account
 - Other account then imports the model
 - Must be in same region
 - Need its ARN, region, and optional KMS key
 - Can be done from the Comprehend console

Amazon Translate



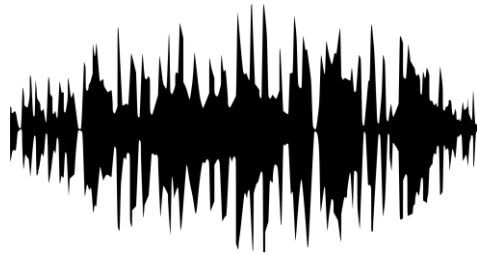
- Natural and accurate **language translation**
- Amazon Translate allows you to **localize content** - such as websites and applications - for **international users**, and to easily translate large volumes of text efficiently.

Source language		Target language
Auto (auto) ▼	↔	French (fr) ▼
Hi my name is Stéphane		Bonjour, je m'appelle Stéphane.
		Portuguese (pt) ▼
		Oi, meu nome é Stéphane.
		Hindi (hi) ▼
		हाय मेरा नाम स्टीफन है

Amazon Transcribe



- Automatically **convert speech to text**
- Uses a **deep learning process** called **automatic speech recognition (ASR)** to convert speech to text quickly and accurately
- **Automatically remove Personally Identifiable Information (PII) using Redaction**
- **Supports Automatic Language Identification** for multi-lingual audio
- Use cases:
 - transcribe customer service calls
 - automate closed captioning and subtitling
 - generate metadata for media assets to create a fully searchable archive



*"Hello my name is Stéphane.
I hope you're enjoying the course!"*

Transcribe – Toxicity Detection

- ML-powered, voice-based toxicity detection capability
- Leverages speech cues: tone and pitch, and text-based cues
- Toxicity categories: sexual harassment, hate speech, threat, abuse, profanity, insult, and graphic....

Transcription preview
You can see the first 5,000 characters of the transcription text below. To download the full text, choose Download full transcript.

Text Audio identification Subtitles **Toxicity**

Toxicity score -- 0.0 to 0.4 0.4 to 0.8 0.8 to 1.0 [Info](#)
Higher score means higher toxicity.

Filters [Info](#) [Reset](#)
Filter out toxic content by increasing threshold values below.

Toxicity Score 0 1
Profanity 0 1
Hate speech 0 1
Sexual 0 1
Insults 0 1

The screenshot displays a web interface for transcription and toxicity detection. It features a 'Transcription preview' section with a 'Download' button. Below this is a 'Toxicity' tab with a 'Toxicity score' legend and a 'Hide filters' button. The main content area shows a transcription of a speech with various segments highlighted in different colors (yellow, orange, red) to indicate toxicity levels. A 'Filters' sidebar on the right allows users to filter out toxic content by adjusting threshold values for Toxicity Score, Profanity, Hate speech, Sexual, and Insults.

Toxicity Categories

Profanity: Speech that contains words, phrases, or acronyms that are impolite, vulgar, or offensive.

Hate speech: Speech that criticizes, insults, denounces, or dehumanizes a person or group on the basis of an identity (such as race, ethnicity, gender, religion, sexual orientation, ability, and national origin).

Sexual: Speech that indicates sexual interest, activity, or arousal using direct or indirect references to body parts, physical traits, or sex.

Insults: Speech that includes demeaning, humiliating, mocking, insulting, or belittling language. This type of language is also labeled as bullying.

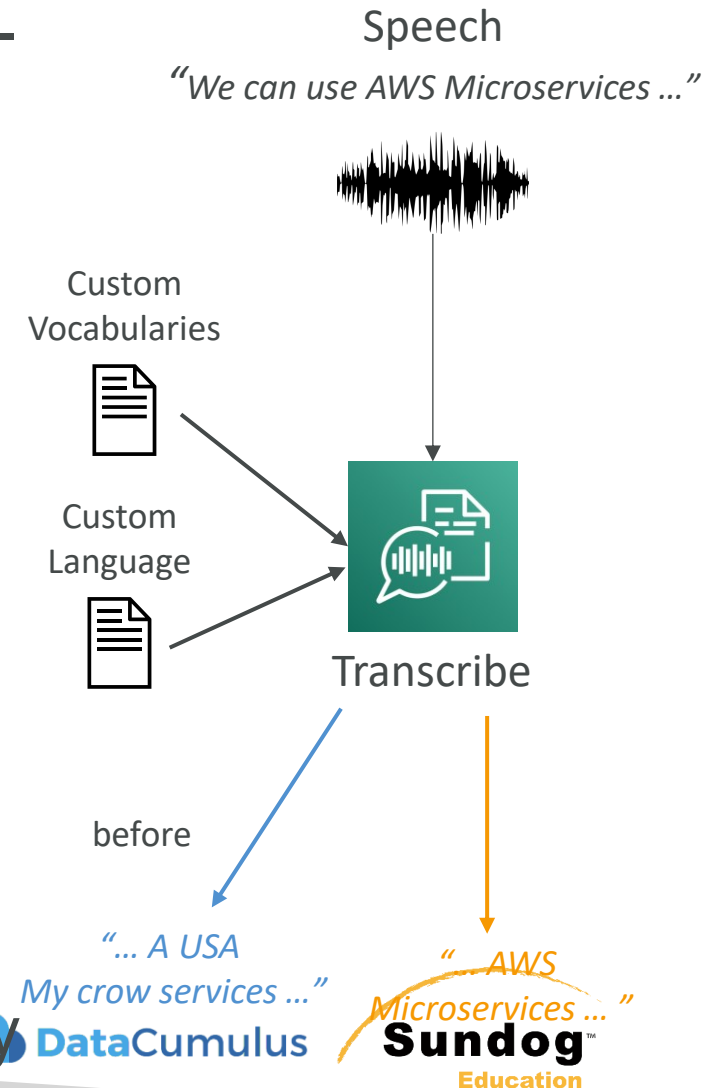
Violence or threat: Speech that includes threats seeking to inflict pain, injury, or hostility toward a person or group.

Graphic: Speech that uses visually descriptive and unpleasantly vivid imagery. This type of language is often intentionally verbose to amplify a recipient's discomfort.

Harassment or abusive: Speech intended to affect the psychological well-being of the recipient, including demeaning and objectifying terms.

Amazon Transcribe – Improving Accuracy

- Allows Transcribe to capture domain-specific or non-standard terms (e.g., technical words, acronyms, jargon...)
- **Custom Vocabularies (for words)**
 - Add specific words, phrases, domain-specific terms
 - Good for brand names, acronyms...
 - Increase recognition of a new word by providing hints (such as pronunciation..)
- **Custom Language Models (for context)**
 - Train Transcribe model on your own domain-specific text data
 - Good for transcribing large volumes of domain-specific speech
 - Learn the context associated with a given word

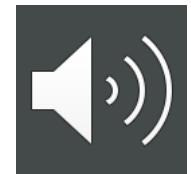


Amazon Polly



- Turn text into lifelike speech using deep learning
- Allowing you to create applications that talk

*Hi! My name is Stéphane
and this is a demo of Amazon Polly*



Polly – Advanced Features

- **Lexicons**

- Define how to read certain specific pieces of text
- AWS => “Amazon Web Services”
- W3C => “World Wide Web Consortium”

- **SSML - Speech Synthesis Markup Language**

- Markup for your text to indicate how to pronounce it
- Example: “Hello, <break> how are you?”

- **Voice engine:** generative, long-form, neural, standard...

- **Speech mark:**

- Encode where a sentence/word starts or ends in the audio
- Helpful for lip-syncing or highlight words as they’re spoken

Action	SSML tag
Adding a pause	<break>
Emphasizing words	<emphasis>
Specifying another language for specific words	<lang>
Placing a custom tag in your text	<mark>
Adding a pause between paragraphs	<p>
Using phonetic pronunciation	<phoneme>
Controlling volume, speaking rate, and pitch	<prosody>
Setting a maximum duration for synthesized speech	<prosody amazon:max-duration>
Adding a pause between sentences	<s>
Controlling how special types of words are spoken	<say-as>
Identifying SSML-enhanced text	<speak>
Pronouncing acronyms and abbreviations	<sub>
Improving pronunciation by specifying parts of speech	<w>
Adding the sound of breathing	<amazon:auto-breaths>
Newscaster speaking style	<amazon:domain name="news">
Adding dynamic range compression	<amazon:effect name="drc">
Speaking softly	<amazon:effect



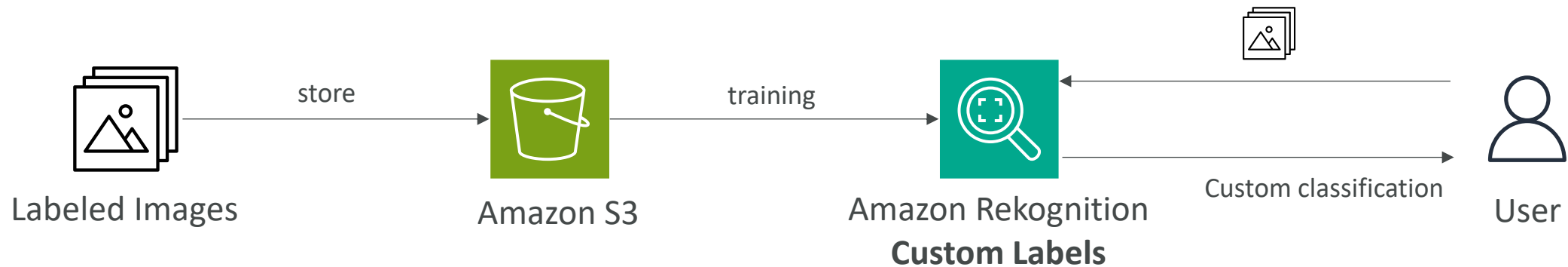
Amazon Rekognition



- Find **objects, people, text, scenes** in **images and videos** using ML
- **Facial analysis** and **facial search** to do user verification, people counting
- Create a database of “familiar faces” or compare against celebrities
- Use cases:
 - Labeling
 - Content Moderation
 - Text Detection
 - Face Detection and Analysis (gender, age range, emotions...)
 - Face Search and Verification
 - Celebrity Recognition
 - **Pathing** (ex: for sports game analysis)

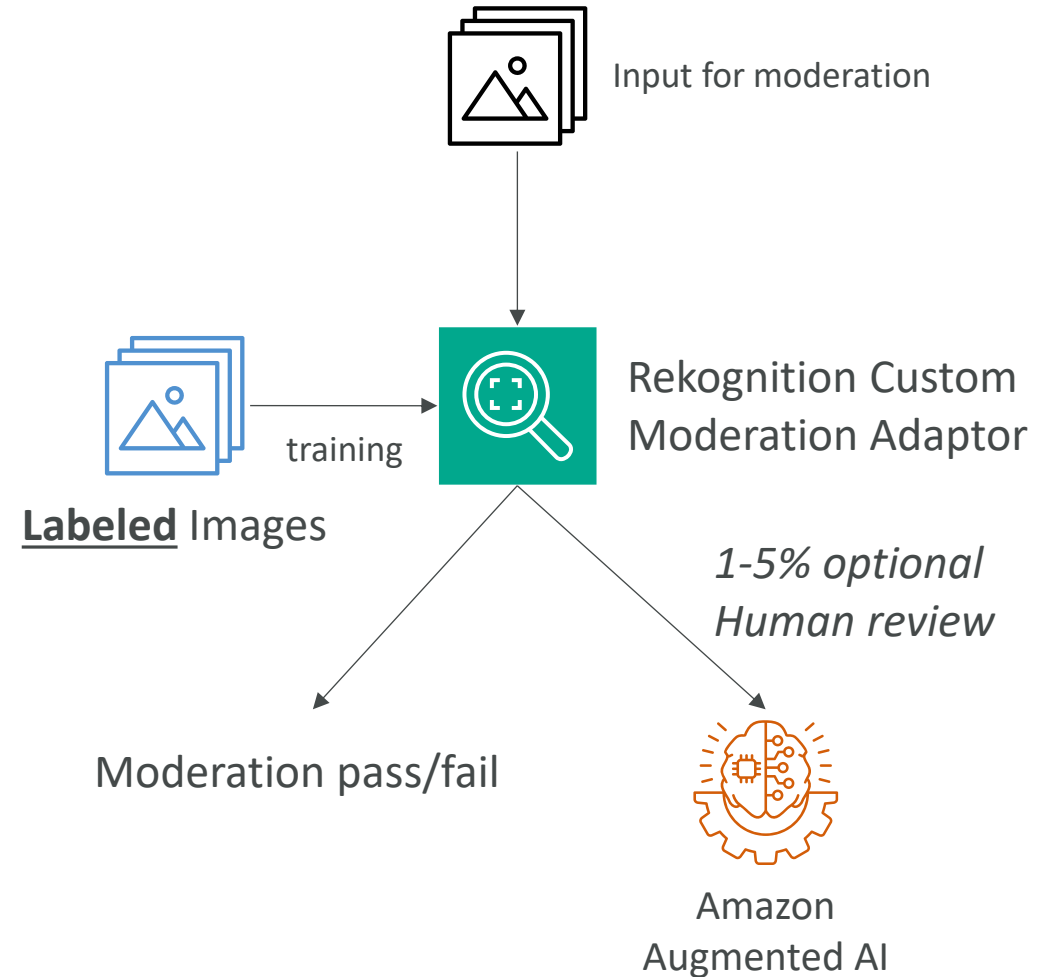
Amazon Rekognition – Custom Labels

- Examples: find your logo in social media posts, identify your products on stores shelves (National Football League – NFL – uses it to find their logo in pictures)
- **Label** your training images and upload them to Amazon Rekognition
- Only needs a few hundred images or less
- Amazon Rekognition creates a custom model on your images set
- New subsequent images will be categorized the custom way you have defined

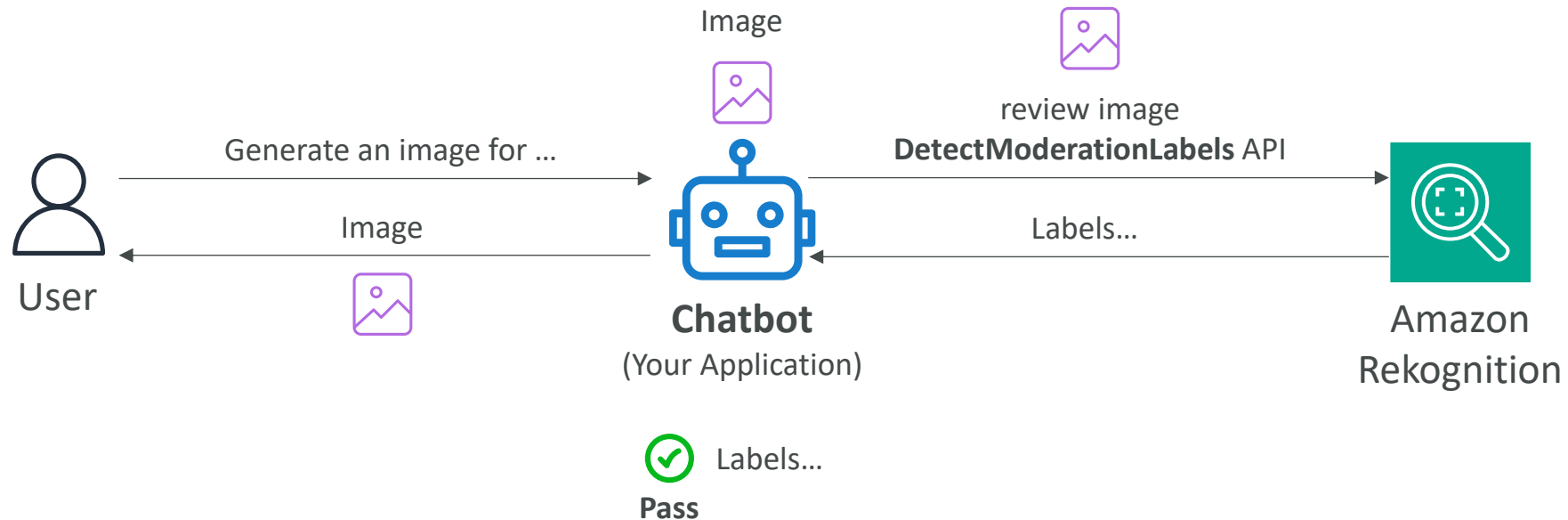


Amazon Rekognition – Content Moderation

- Automatically detect inappropriate, unwanted, or offensive content
- Example: filter out harmful images in social media, broadcast media, advertising...
- Bring down human review to 1-5% of total content volume
- Integrated with Amazon Augmented AI (Amazon A2I) for human review
- **Custom Moderation Adaptors**
 - Extends Rekognition capabilities by providing your own **labeled** set of images
 - Enhances the accuracy of Content Moderation or create a specific use case of Moderation



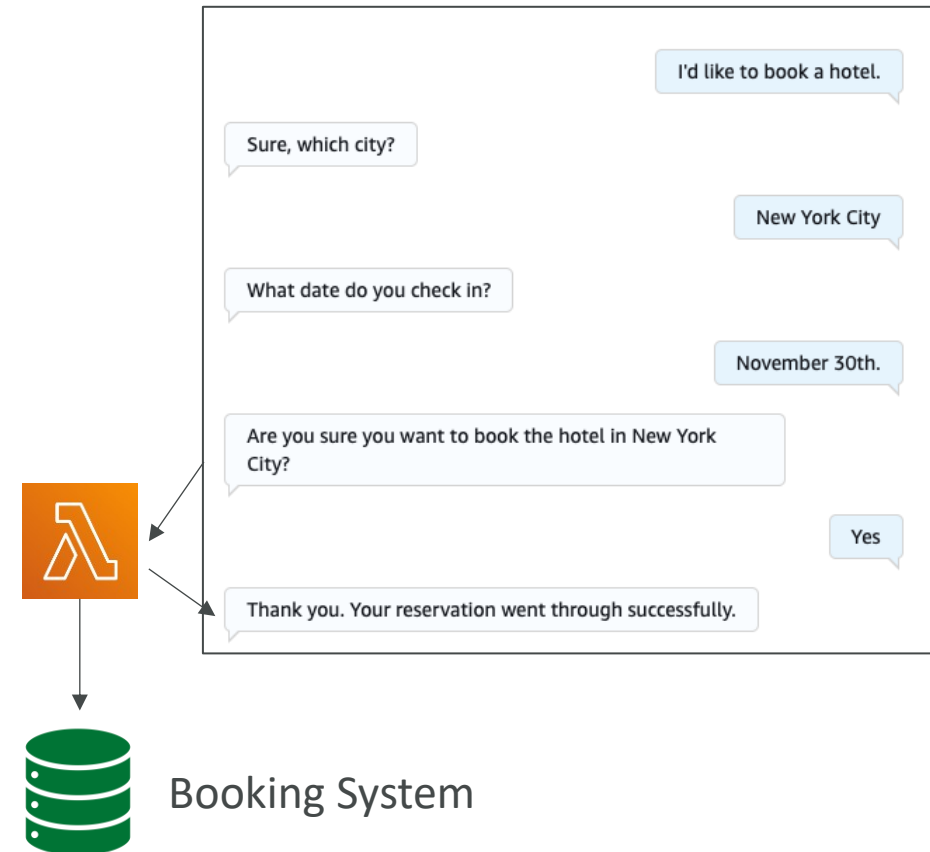
Content Moderation API – Diagram



Amazon Lex



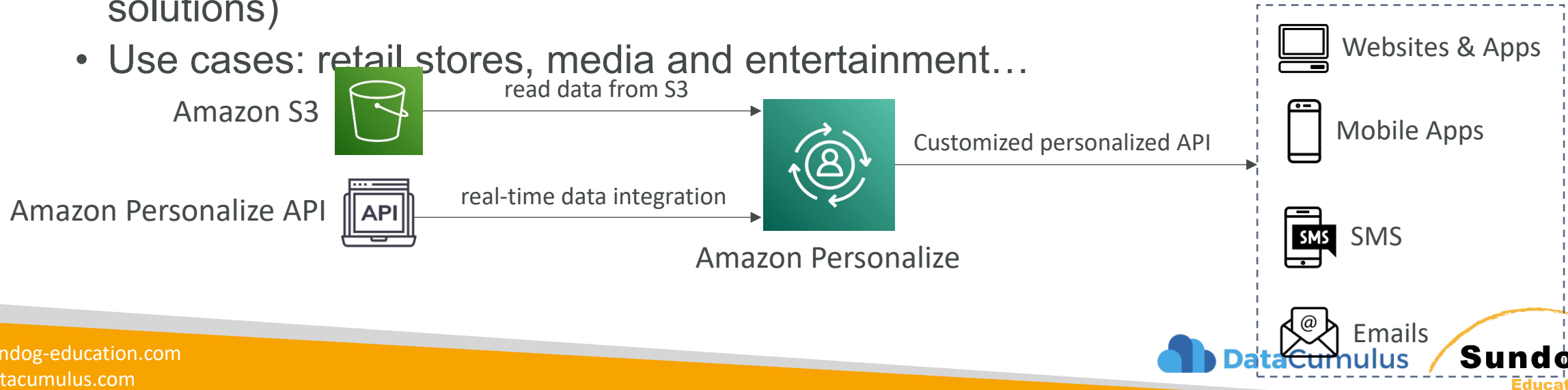
- Build chatbots quickly for your applications using voice and text
- Example: a chatbot that allows your customers to order pizzas or book a hotel
- Supports multiple languages
- Integration with AWS Lambda, Connect, Comprehend, Kendra
- The bot automatically understands the user intent to invoke the correct Lambda function to “fulfill the intent”
- The bot will ask for “Slots” (input parameters) if necessary



Amazon Personalize



- Fully managed ML-service to build apps with real-time personalized recommendations
- Example: personalized product recommendations/re-ranking, customized direct marketing
 - Example: User bought gardening tools, provide recommendations on the next one to buy
- Same technology used by Amazon.com
- Integrates into existing websites, applications, SMS, email marketing systems, ...
- Implement in days, not months (you don't need to build, train, and deploy ML solutions)
- Use cases: retail stores, media and entertainment...



Amazon Personalize – Recipes

- Algorithms that are prepared for specific use cases
- You must provide the training configuration on top of the recipe
- Example recipes:
 - Recommending items for users (**USER_PERSONALIZATION** recipes)
 - User-Personalization-v2
 - Ranking items for a user (**PERSONALIZED_RANKING** recipes)
 - Personalized-Ranking-v2
 - Recommending trending or popular items (**POPULAR_ITEMS** recipes)
 - Trending-Now, Popularity-Count
 - Recommending similar items (**RELATED_ITEMS** recipes)
 - Similar-Items
 - Recommending the next best action (**PERSONALIZED_ACTIONS** recipes)
 - Next-Best-Action
 - Getting user segments (**USER_SEGMENTATION** recipes)
 - Item-Affinity
- **NOTE:** recipes and personalize are for recommendations

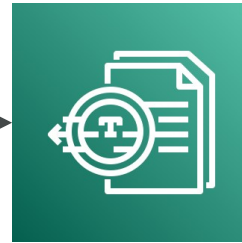
Amazon Textract



- Automatically extracts text, handwriting, and data from any scanned documents using AI and ML



analyze



Amazon Textract

result

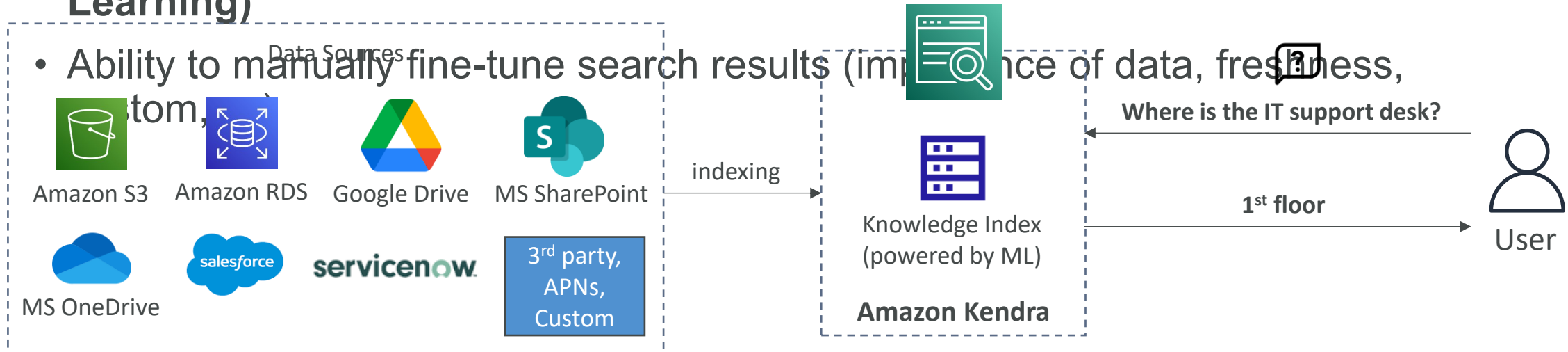
```
{  
  "Document ID": "123456789-005",  
  "Name": "",  
  "SEX": "F",  
  "DOB": "23.05.1997",  
  ...  
}
```

- Extract data from forms and tables
- Read and process any type of document (PDFs, images, ...)
- Use cases:
 - Financial Services (e.g., invoices, financial reports)
 - Healthcare (e.g., medical records, insurance claims)
 - Public Sector (e.g., tax forms, ID documents, passports)

Amazon Kendra

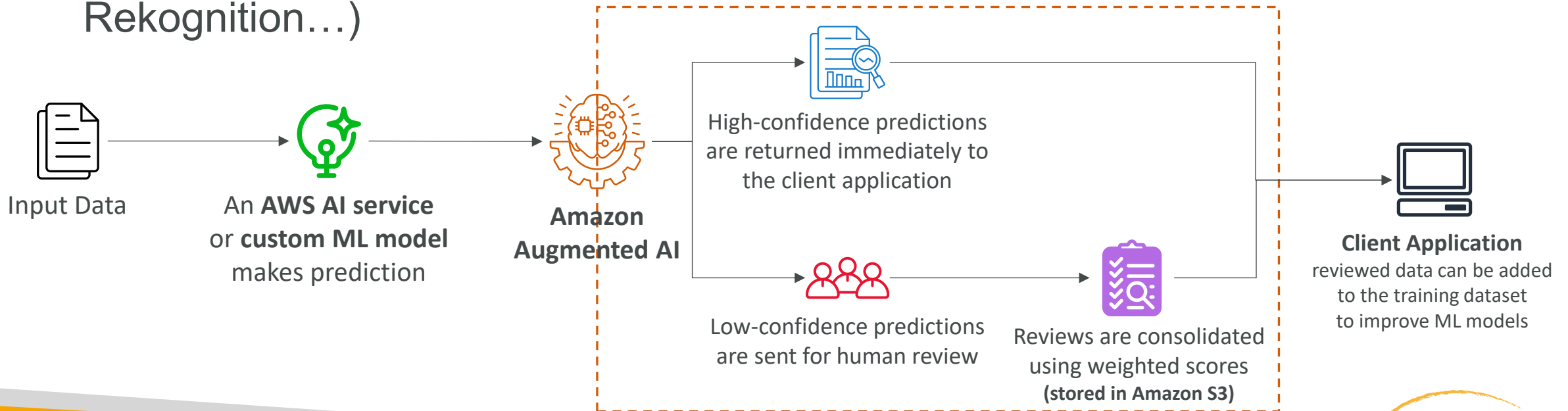


- Fully managed **document search service** powered by Machine Learning
- Extract answers from within a document (text, pdf, HTML, PowerPoint, MS Word, FAQs...)
- Natural language search capabilities
- Learn from user interactions/feedback to promote preferred results (**Incremental Learning**)

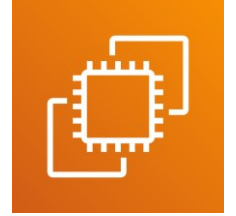


Amazon Augmented AI (A2I)

- Human oversight of Machine Learning predictions in production
 - Can be your own employees, over 500,000 contractors from AWS, or AWS Mechanical Turk
 - Some vendors are pre-screened for confidentiality requirements
- The ML model can be built on AWS or elsewhere (SageMaker, Rekognition...)



Amazon EC2



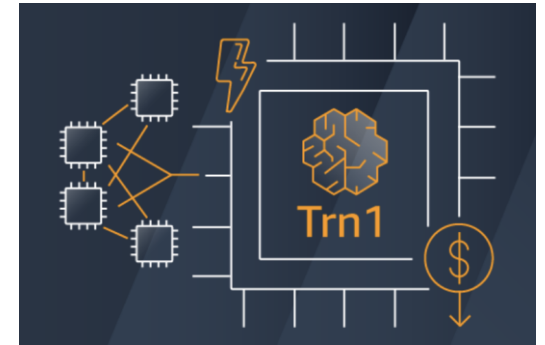
- EC2 is one of the most popular of AWS' offerings
- EC2 = Elastic Compute Cloud = Infrastructure as a Service
- It mainly consists in the capability of :
 - Renting virtual machines (EC2)
 - Storing data on virtual drives (EBS)
 - Distributing load across machines (ELB)
 - Scaling the services using an auto-scaling group (ASG)
- Knowing EC2 is fundamental to understand how the Cloud works

EC2 sizing & configuration options

- Operating System (**OS**): Linux, Windows or Mac OS
- How much compute power & cores (**CPU**)
- How much random-access memory (**RAM**)
- How much storage space:
 - Network-attached (**EBS & EFS**)
 - hardware (**EC2 Instance Store**)
- Network card: speed of the card, Public IP address
- Firewall rules: **security group**
- Bootstrap script (configure at first launch): EC2 User Data

Amazon's Hardware for AI

- GPU-based EC2 Instances (P3, P4, P5..., G3...G6...)
- **AWS Trainium**
 - ML chip built to perform Deep Learning on 100B+ parameter models
 - Trn1 instance has for example 16 Trainium Accelerators
 - 50% cost reduction when training a model
- **AWS Inferentia**
 - ML chip built to deliver inference at high performance and low cost
 - Inf1, Inf2 instances are powered by AWS Inferentia
 - Up to 4x throughput and 70% cost reduction



Amazon Lookout: Anomaly Detection

- Uses ML to detect and diagnose anomalies
- Identify unusual variances
 - Business performance
 - Ad campaigns
 - Whatever
- **Detectors** monitor **datasets** to find **anomalies**
- The thing you're optimizing is the **measure**, the features that influence it are **dimensions**.
- Specialized version for industrial use
 - For equipment: integrate with equipment sensors for predictive maintenance
 - For vision: automate quality inspection with computer vision



Amazon Lookout
for Equipment

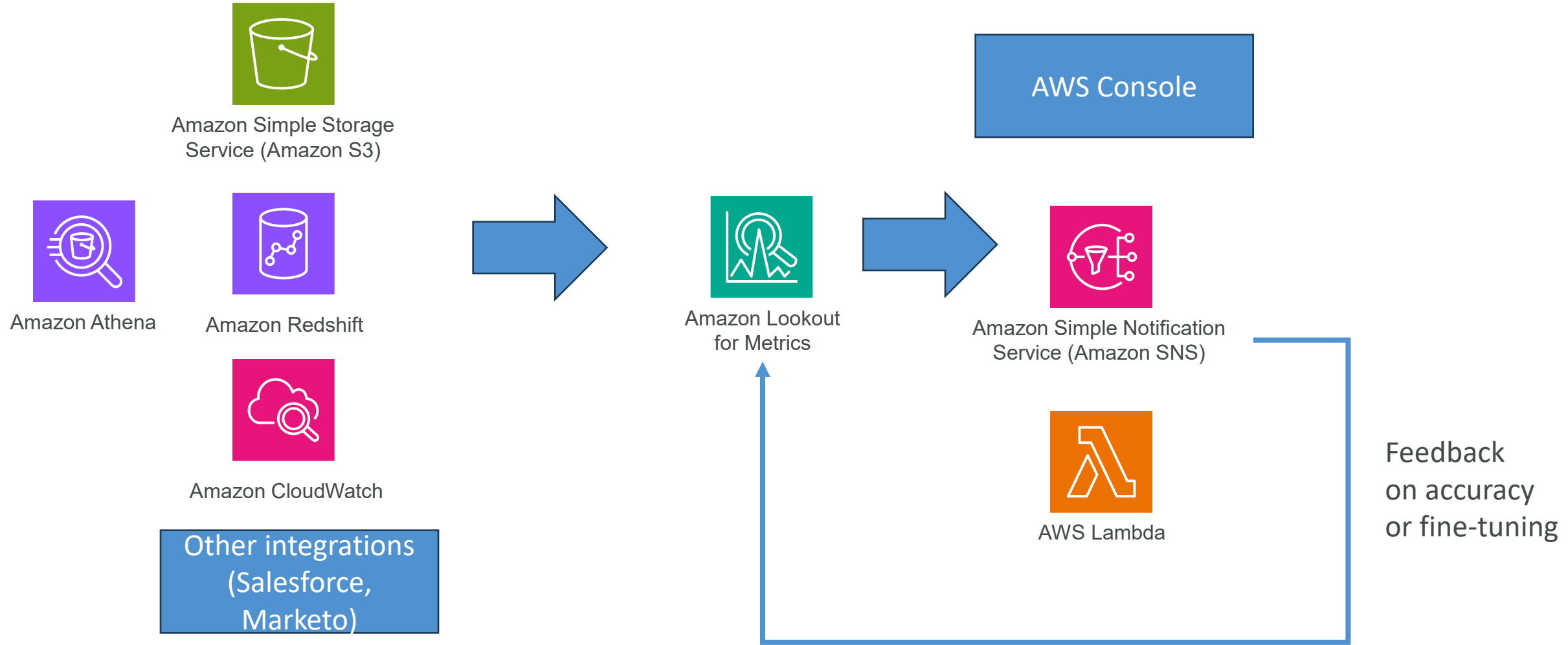


Amazon Lookout
for Metrics



Amazon Lookout
for Vision

Lookout for Metrics: (Slightly) More Depth

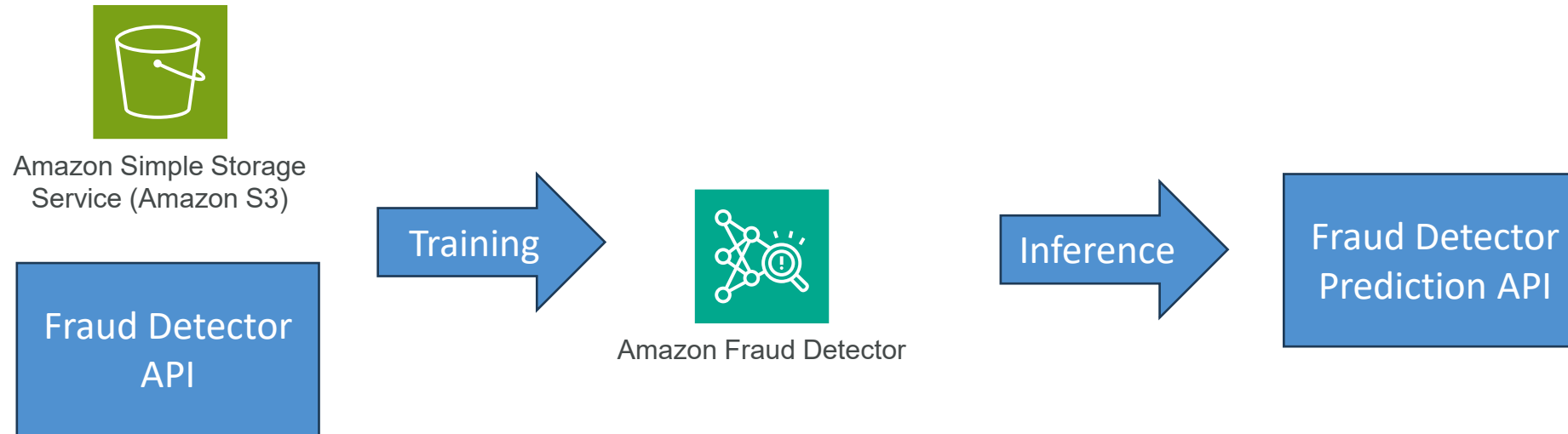


Amazon Fraud Detector

- Does what it says on the tin
- Examples of fraud:
 - Online payments, new accounts, trial program abuse, account takeovers
- Fully managed ML model customized to your data
 - Automatic model creation
 - Continuous learning over time
- Insights into the importance of your features (model variables)
- Ingests data from S3 or an API
- Rule-based actions
- SageMaker integration



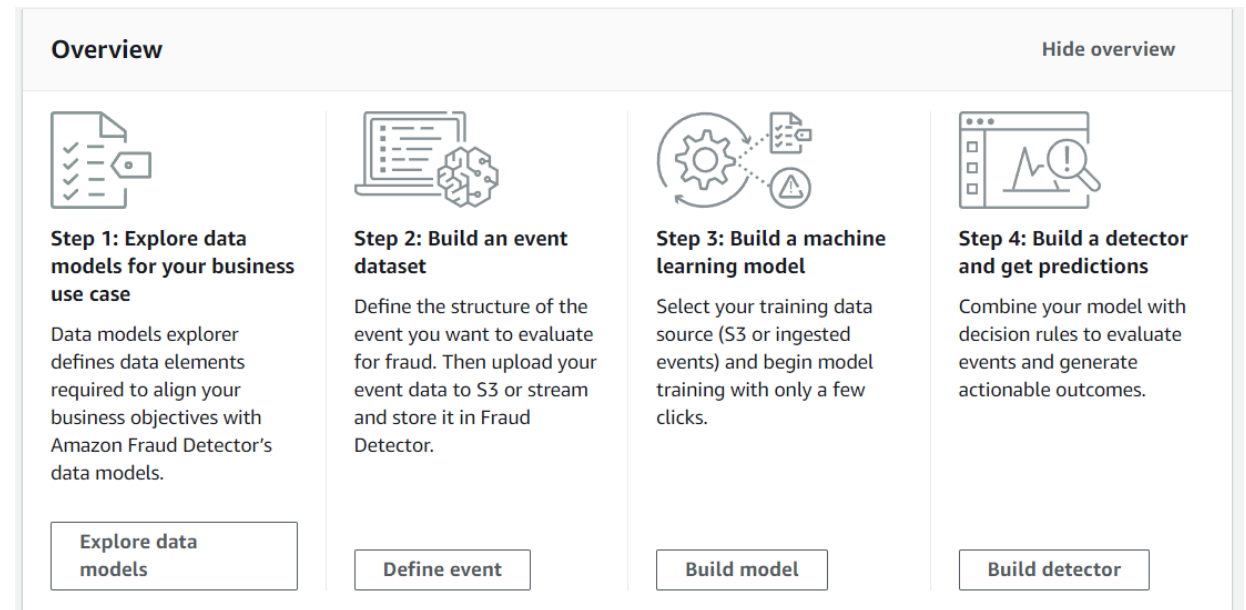
Amazon Fraud Detector



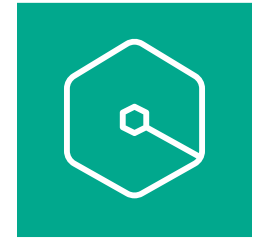
ETL, feature engineering,
model selection, training,
Tuning, validation, and deployment

Using Amazon Fraud Detector

- Choose a business use case (account fraud, etc.)
- Create an event type to monitor, and entities associated with that event
 - Point this to the data source for this event
 - Define labels
 - Create necessary IAM permissions
- Create model
- Train model
- Review performance
- Deploy



Amazon Q Business



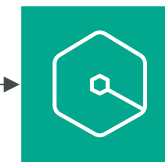
- Fully managed Gen-AI assistant for your employees
- Based on your company's knowledge and data
 - Answer questions, provide summaries, generate content, automate tasks
 - Perform routine actions (e.g., submit time-off requests, send meeting invites)
- Built on Amazon Bedrock (but you can't choose the underlying FM)

What you can ask Amazon Q Business?



Employee

- Write a job posting for a Senior Product Marketing Manager role...
- Create a social media post under 50 words to advertise the new role...
- What was discussed during the team meetings in the week of 4/12?

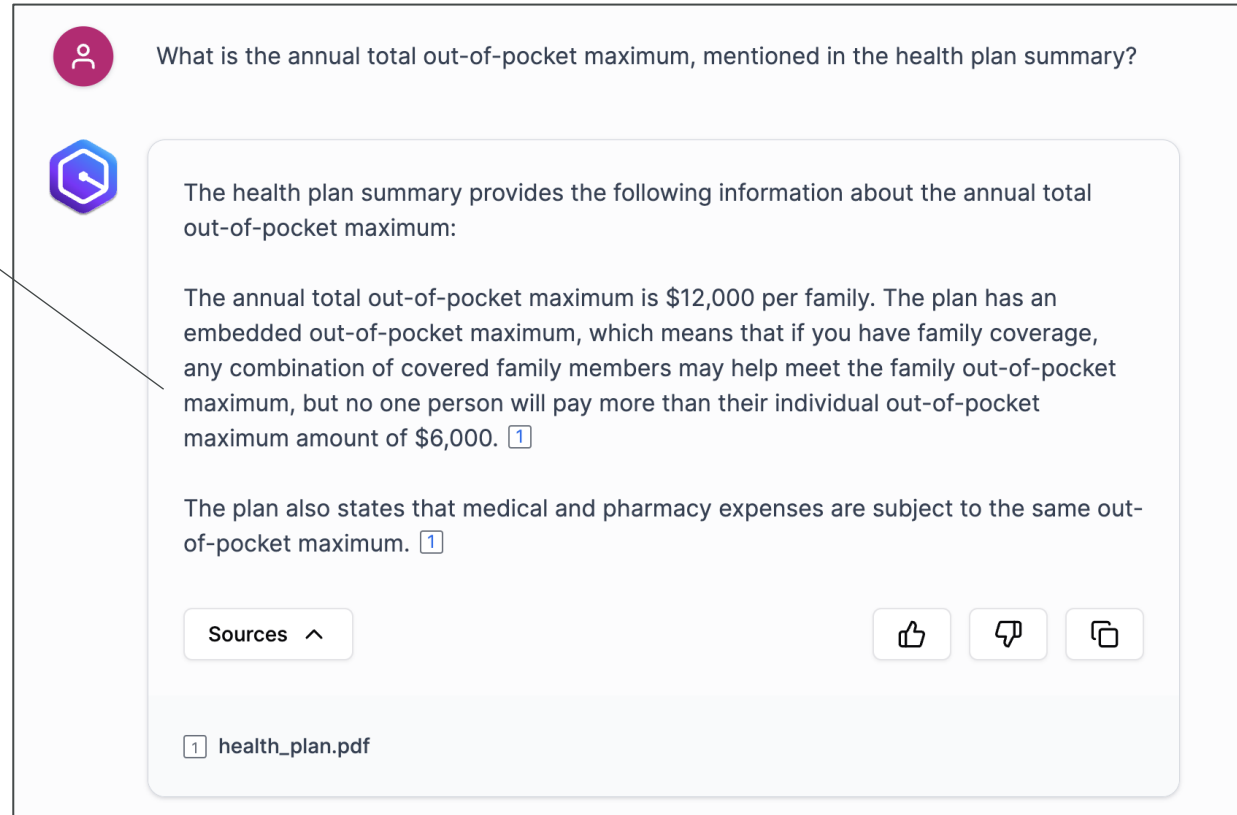
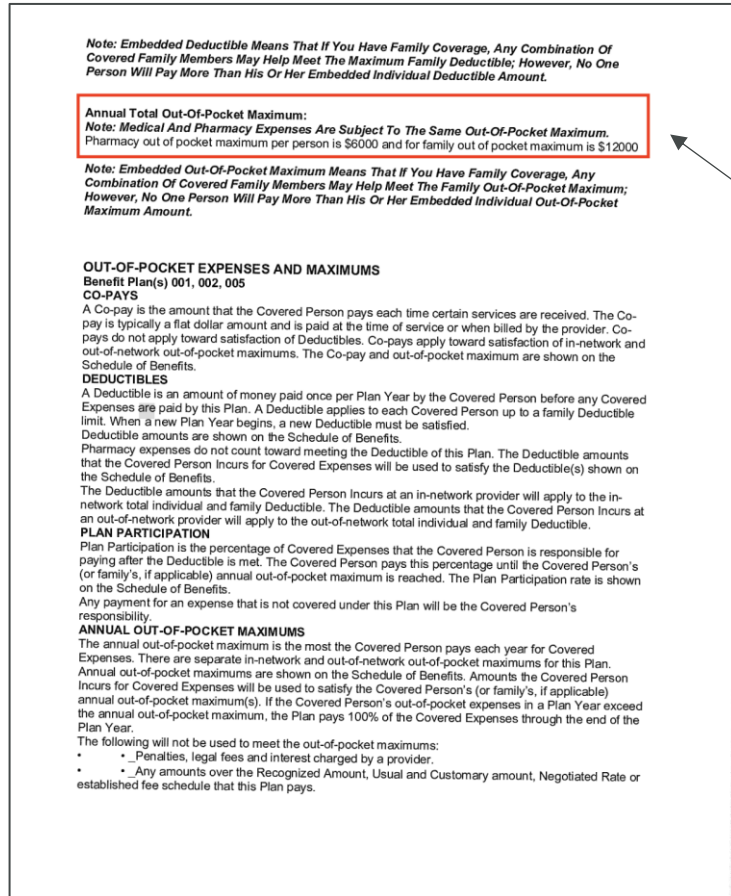


Amazon Q Business

Company's
Internal Data



Amazon Q Business Example

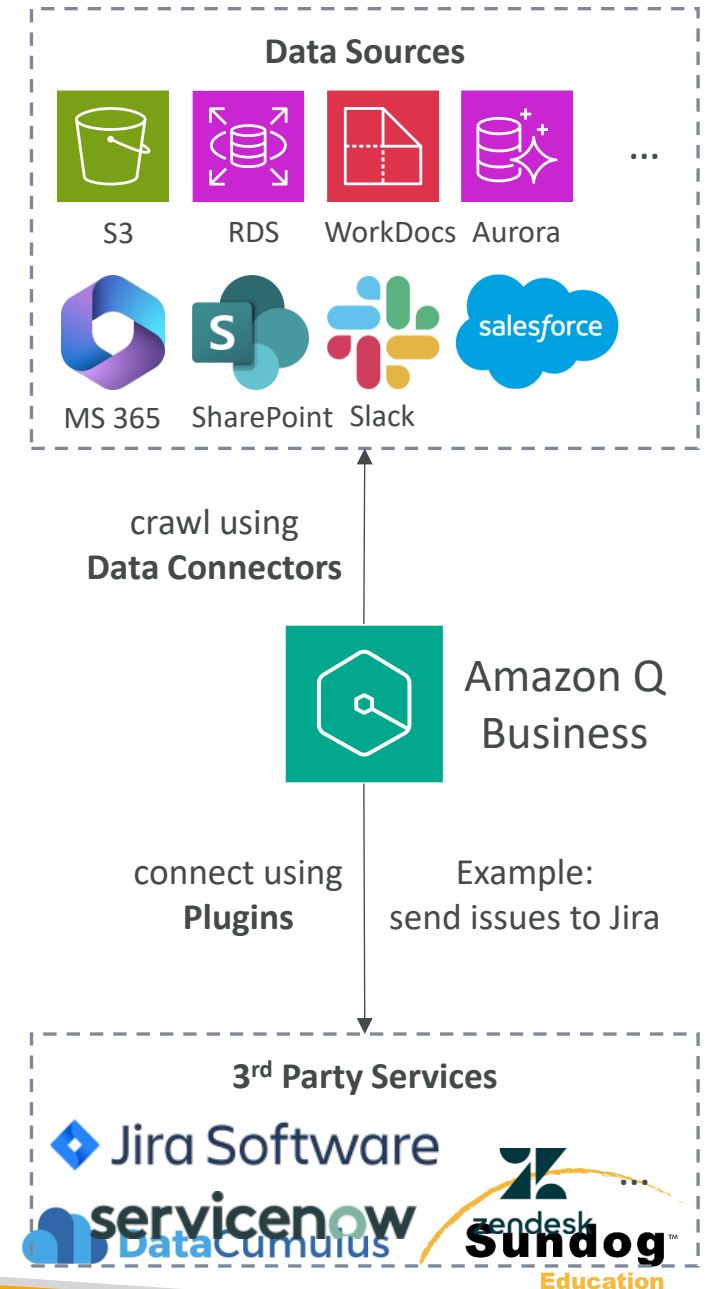


Company Document



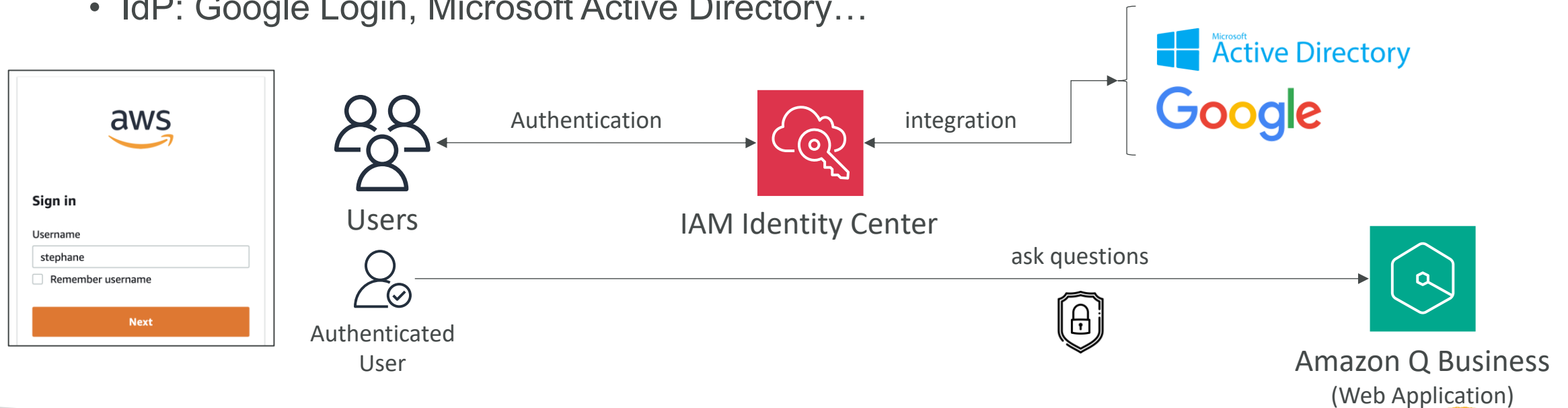
Amazon Q Business

- **Data Connectors (fully managed RAG)** – connects to 40+ popular enterprise data sources
 - Amazon S3, RDS, Aurora, WorkDocs...
 - Microsoft 365, Salesforce, GDrive, Gmail, Slack, Sharepoint...
- **Plugins** – allows you to interact with 3rd party services
 - Jira, ServiceNow, Zendesk, Salesforce...
 - **Custom Plugins** – connects to any 3rd party application using APIs



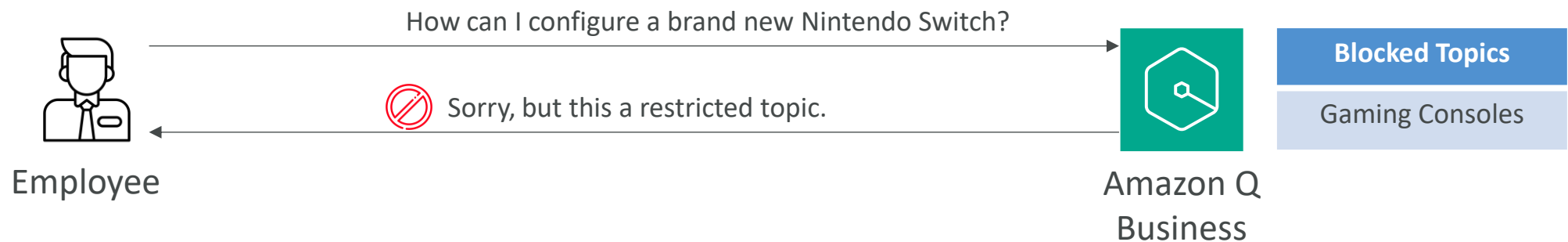
Amazon Q Business + IAM Identity Center

- Users must be authenticated through **IAM Identity Center**
- Users receive responses generated only from the documents they have access to
- IAM Identity Center can be configured with external Identity Providers
 - IdP: Google Login, Microsoft Active Directory...

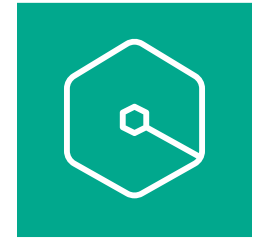


Amazon Q Business – Admin Controls

- Controls and customize responses to your organizational needs
- Admin controls == Guardrails
- Block specific words or topics
- Respond only with internal information (vs using external knowledge)
- Global controls & topic-level controls (more granular rules)



Amazon Q Apps (Q Business)



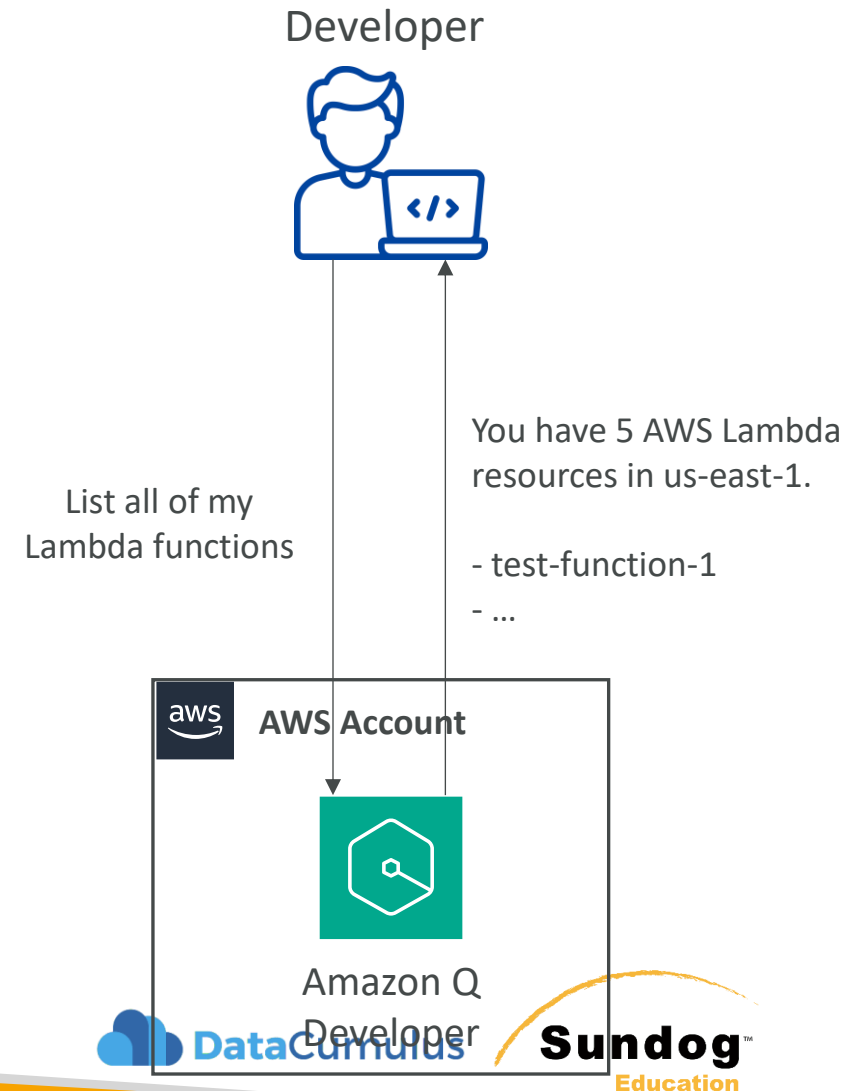
- Create Gen AI-powered apps without coding by using natural language
- Leverages your company's internal data
- Possibility to leverage plugins (Jira, etc...)

The image shows a two-part interface for Amazon Q Apps. The left panel is the 'Amazon Q Apps Creator' dashboard, titled 'Your generative AI productivity app generator'. It features a sidebar with 'Chat', 'Apps', and 'Library' tabs. The main area contains a prompt: 'Tired of repetitive tasks? Tell me what you need done and I'll create a custom app tailored for your needs.' Below this is a text input field with a sample prompt about being a professional editor, a character count of 427 / 10000, and 'Skip this step' and 'Generate' buttons. At the bottom, there are four example app cards: 'Content Creator', 'Interview Question Generator', 'Meeting Notes Summarizer', and 'Grammar Corrector'. The right panel shows the 'Document Editing Assistant' app interface. It has a sidebar with 'Apps' and 'Library' tabs. The main area displays the app's name and description: 'Reviews and suggests corrections for documents to improve grammar, spelling, style, and tone consistency'. Below this is an 'Upload Document' section with a file upload area and a 'Browse for files' button. At the bottom, there is a 'Text output' section titled 'Edit Suggestions' and a 'Run' button with a 'Reset' button next to it.

<https://aws.amazon.com/blogs/aws/amazon-q-business-now-generally-available-helps-boost-workforce-productivity-with-generative-ai/>

Amazon Q Developer

- Answer questions about the AWS documentation and AWS service selection
- Answer questions about resources in your AWS account
- Suggest CLI (Command Line Interface) to run to make changes to your account
- Helps you do bill analysis, resolve errors, troubleshooting...



Amazon Q Developer

Amazon Q

Change the timeout for Lambda function test-api-1 in the Singapore region, to 10 seconds.

To change the timeout for your Lambda function test-api-1 in the Singapore region to 10 seconds, you can use the AWS CLI. Run the following command:

```
aws lambda update-function-configuration --function-name test-api-1 --region ap-southeast-1 --timeout 10
```

This will update the configuration of the specified Lambda function to set the timeout to 10 seconds.

Some things to keep in mind regarding Lambda timeouts:

- The default timeout is 3 seconds, but it can be configured between 1-15 minutes as needed.
- Factors like large data processing, network latency from calls to external services, complex computations can cause functions to timeout.
- You should choose the timeout carefully based on expected maximum function duration. Setting it too low risks unexpected timeouts, too high wastes compute resources.
- Tools like Step Functions allow building applications with tasks that can run for extended periods by coordinating multiple Lambdas.
- If timeouts occur, review function logic and use profiling tools to identify bottlenecks before increasing timeout.

Let me know if you need any clarification or have additional questions!

Ask me anything about AWS

Max 1000 characters

Use of Amazon Q is subject to the [AWS Responsible AI Policy](#)

Understand and manage
your cloud infrastructure

Amazon Q

What were the top three highest-cost services in Q1?

The top 3 highest-cost AWS services in Q1 2024 were:

1. Amazon SageMaker - \$ (29.88% of total cost)
2. Amazon Elastic Container Service - \$ (17.97% of total cost)
3. AWS Config - \$ (8.13% of total cost)

The following Cost Explorer parameters were used to answer your question:

- Date range: 2024-01-01 through 2024-03-31 (inclusive)
- Granularity: "MONTHLY"
- Metrics: "UnblendedCost"
- Group by: "SERVICE"
- Filter: Excludes RECORD_TYPE "Credit", "Refund"

Amazon Q's cost analysis capability is in preview and can make mistakes. Please verify your cost data with AWS Cost Explorer. Use the thumb icon to provide feedback and help us improve.

Sources

[1] [Visualize this data in AWS Cost Explorer](#)

[2] [AWS Cost Explorer GetCostAndUsage API Reference](#)

[3] [Modifying your chart - AWS Cost Management](#)

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📄

New conversation

Ask me anything about AWS

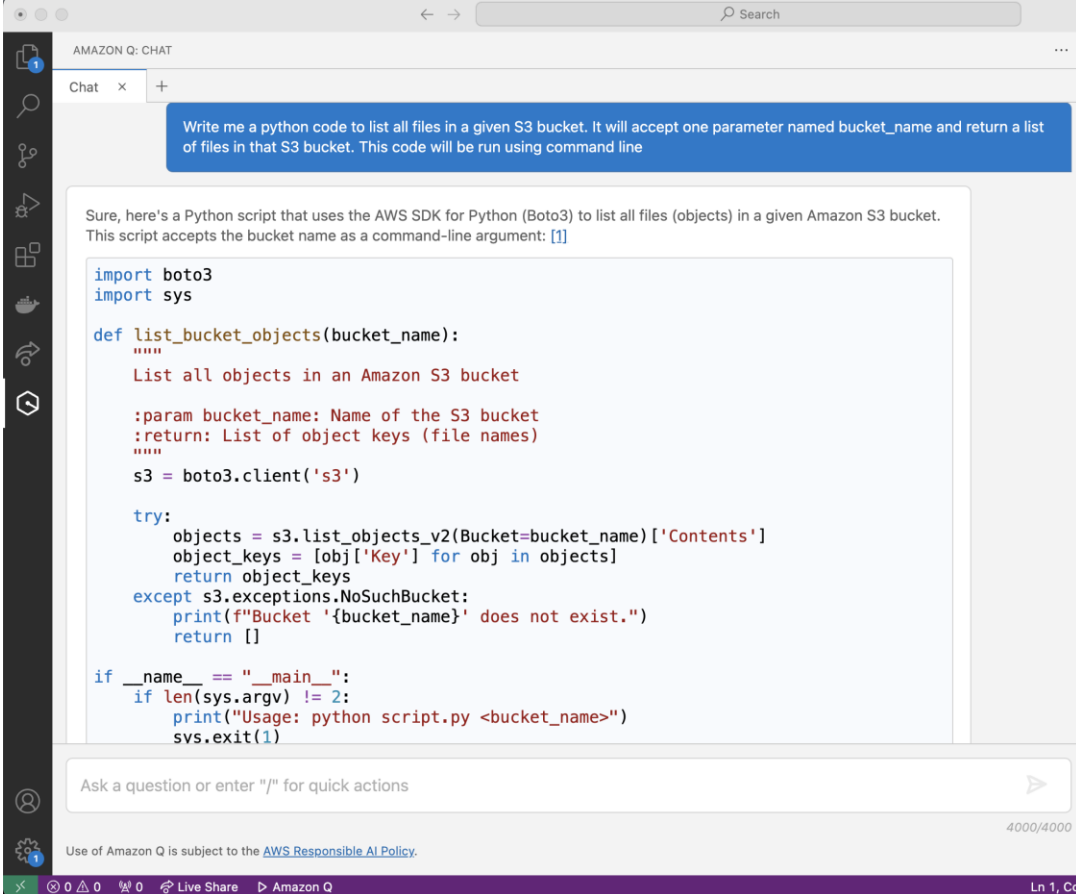
Max 1000 characters

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Understand your AWS costs

Amazon Q Developer

- AI code companion to help you code new applications (similar to GitHub Copilot)
- Supports many languages: Java, JavaScript, Python, TypeScript, C#...
- Real-time code suggestions and security scans
- Software agent to implement features, generate documentation, bootstrapping new projects



The screenshot shows the Amazon Q Developer chat interface. At the top, a blue prompt bar contains the text: "Write me a python code to list all files in a given S3 bucket. It will accept one parameter named bucket_name and return a list of files in that S3 bucket. This code will be run using command line". Below this, the chat response area shows a message from the AI: "Sure, here's a Python script that uses the AWS SDK for Python (Boto3) to list all files (objects) in a given Amazon S3 bucket. This script accepts the bucket name as a command-line argument: [1]". The response includes a Python code block with the following content:

```
import boto3
import sys

def list_bucket_objects(bucket_name):
    """
    List all objects in an Amazon S3 bucket

    :param bucket_name: Name of the S3 bucket
    :return: List of object keys (file names)
    """
    s3 = boto3.client('s3')

    try:
        objects = s3.list_objects_v2(Bucket=bucket_name)['Contents']
        object_keys = [obj['Key'] for obj in objects]
        return object_keys
    except s3.exceptions.NoSuchBucket:
        print(f"Bucket '{bucket_name}' does not exist.")
        return []

if __name__ == "__main__":
    if len(sys.argv) != 2:
        print("Usage: python script.py <bucket_name>")
        sys.exit(1)
```

At the bottom of the chat window, there is an input field with the placeholder text "Ask a question or enter '/' for quick actions" and a send button. The bottom status bar of the application shows "Use of Amazon Q is subject to the [AWS Responsible AI Policy](#)." and a "Live Share" button.

Amazon Q Developer – IDE Extensions

- Integrates with IDE (Integrated Development Environment) to help with your software development needs
 - Answer questions about AWS development
 - Code completions and code generation
 - Scan your code for security vulnerabilities
 - Debugging, optimizations, improvements



Visual Studio Code



Visual Studio

